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3.2 | Strut to Tree Connection

Use Simpson Strong Tie online selection tool.

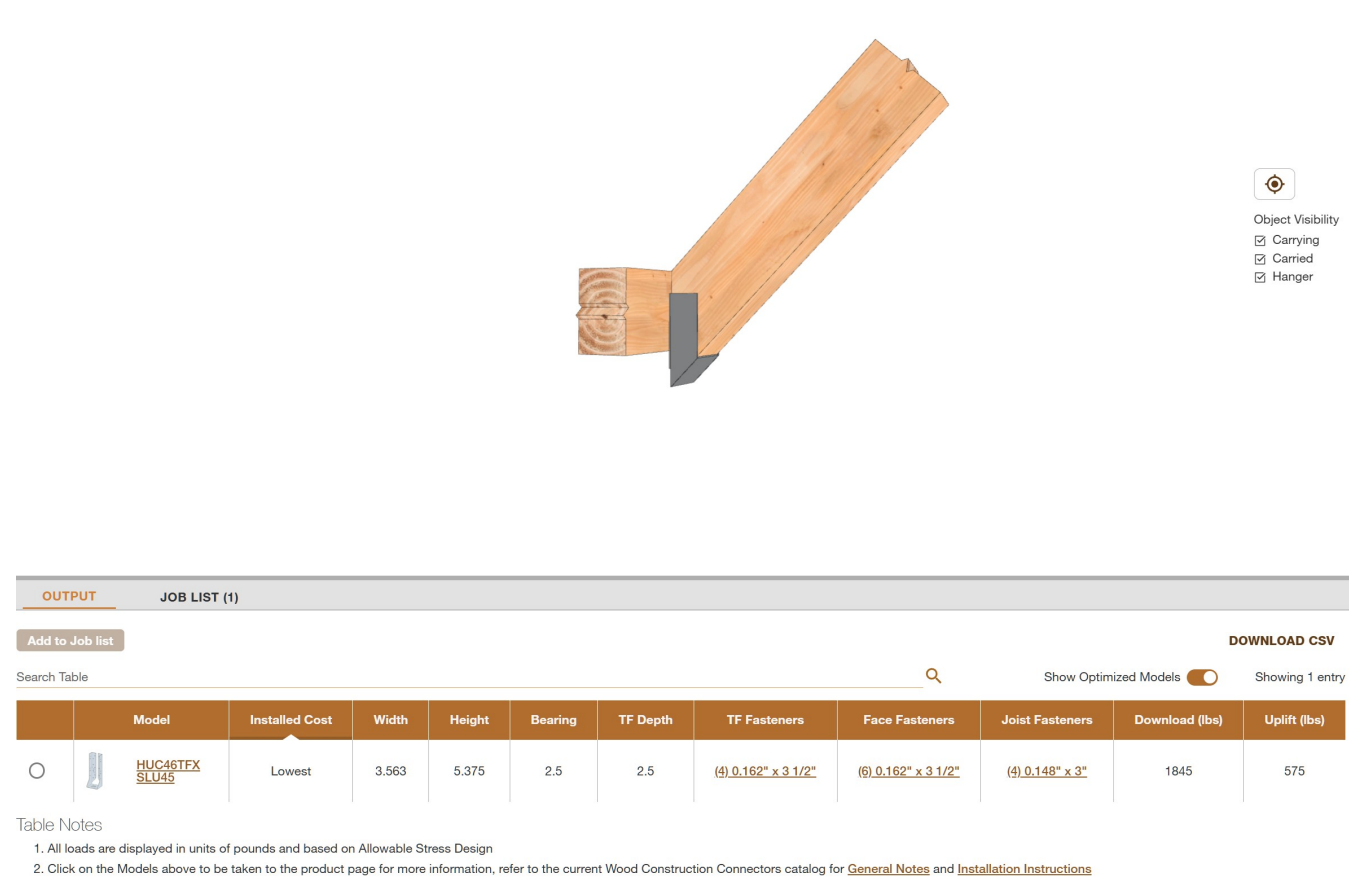


Fig. 1 - Screenshot: Option 1 | time: no time



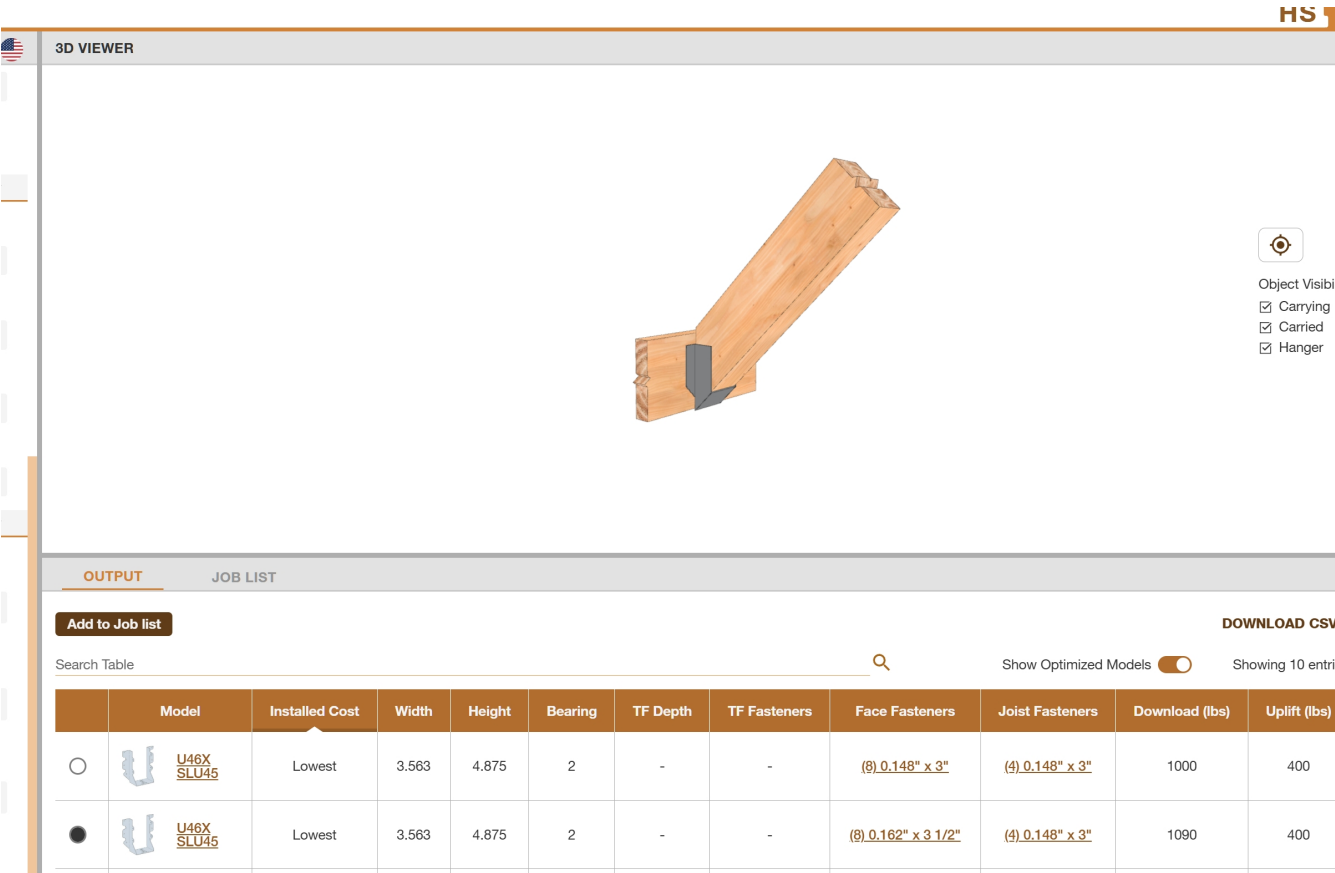


Fig. 2 - Screenshot: Option 2 | time: no time

3.2 - 2 | Top rail Corner

Use AWC online connection tool.

← ConnectionCalc Results	
Analysis Type	
Design Method:	Allowable Stress
Connection Loading:	Lateral
Fastener Type:	Wood Screw
Main Member Parameters	
Main Member material category:	Solid Lumber/Timber
Type:	Douglas Fir-Larch
Main Member Thickness:	1-1/2 in
Load to Grain Angle:	0 deg
Side Member Parameters	
Side Member material category:	Steel
Side Member Thickness:	14 gage
Wood Screw Parameters	
Size:	No.8
Length:	1-1/2 in
Analysis Factors	
Load Duration (Cd):	1
Wet Service (CM):	0.7
End Grain (Ceg):	1
Temperature Factor (Ct):	0.8
Results	
Adjusted ASD Capacity:	55 lb.
Yield Modes	

Fig. 3 - Screenshot: Top Rail - Corner Plate Input



← ConnectionCalc Results	
②	1
Temperature Factor (C):	
②	0.8
Results	
Adjusted ASD Capacity:	55 lb.
Yield Modes	
Z' Im:	
②	195 lb.
Z' Is:	
②	156 lb.
Z' Il:	
②	78 lb.
Z' Illm:	
②	89 lb.
Z' Ills:	
②	55 lb.
Z' IV:	
②	72 lb.
Notes	
Douglas Fir- Larch Main Member Dowel Bearing Strength (psi):	
②	4,650 psi
Side Member Assumed Steel Alloy:	
②	ASTM A653 G33
Steel Side Member Dowel Bearing Strength (psi):	
②	61,850 psi
Fastener Bending Yield Strength:	
②	90,000 psi
Tip length:	
②	2x diameter
Screw Dowel Bending Strength rounding:	
②	nearest 50 psi
Lateral Results Rounding:	
②	nearest 10 lb
Toe-nails in wood members:	
②	results do not apply
Disclaimer:	
While every effort has been made to insure the accuracy of the information presented, and special effort has been made to assure that the information reflects the state-of-the-art, neither the American Wood Council nor its members assume any responsibility for any particular design prepared from this Connection Calculator. Those using this Connection Calculator assume all liability from its use.	

Fig. 4 - Screenshot: Top Rail - Corner Plate Capacity

Use 4-#8 screws = 55 lbs * 4 = Capacity 220 lbs | Demand = 200 lbs.

